

ACTL2102 Tutorial

Week 4: NCD Systems, Time in Transient States, and the Exponential Distribution

Nick Guo

School of Risk and Actuarial Studies, UNSW Sydney

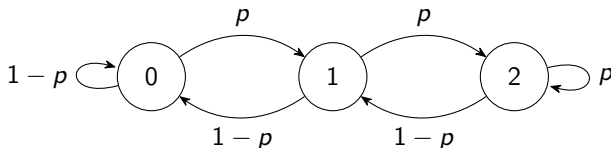
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Plan

- ▶ Recap: NCD systems, mean time spent in transient states, the exponential distribution, counting processes and the Poisson process
- ▶ Main questions: **2.19, 2.18, 3.1, 3.3, 3.4, 3.12**

NCD as a Markov Chain

In a no claim discount (NCD) scheme the premium depends on the policyholder's claim record. The discount level is modelled as a Markov chain: a claim-free year moves the policyholder up one level, and a year with one or more claims moves them down.



Here p is the probability of a claim-free year. At the highest level a claim-free year leaves the level unchanged, and at the lowest level a year with a claim does, which gives the two self-loops.

Schemes differ in the number of levels lost after a claim and in whether p depends on the current level, so $\mathbf{P}$ is built from the rules as stated.

Example: a Three-Level NCD Scheme

Levels 0, 1, 2 carry discounts of 0%, 25%, 50%. A claim-free year (probability 0.9) moves the policyholder up one level and a year with a claim moves them down one, where such a level exists.

$$\mathbf{P} = \begin{pmatrix} 0.1 & 0.9 & 0 \\ 0.1 & 0 & 0.9 \\ 0 & 0.1 & 0.9 \end{pmatrix}$$

The chain is finite, irreducible and aperiodic, so the stationary distribution is unique and is the limiting distribution. From $\boldsymbol{\pi} = \boldsymbol{\pi}\mathbf{P}$, $\pi_1 = 9\pi_0$ and $\pi_2 = 81\pi_0$, and $\sum_j \pi_j = 1$ gives

$$\boldsymbol{\pi} = \left(\frac{1}{91}, \frac{9}{91}, \frac{81}{91} \right) \approx (0.011, 0.099, 0.890).$$

About 89% of policyholders are at the highest discount level in the long run. The mean times between visits, $m_{jj} = 1/\pi_j$, are $m_{00} = 91$, $m_{11} = 10.11$ and $m_{22} = 1.12$ years.

Exercise 2.19: Five-level NCD scheme: classification, long run, recurrence time

Exercise 2.19: [dmc18, Solution] In car insurance no claim discount schemes are often used. A car insurer has five no-claim levels, $\{-1, 0, 1, 2, 3\}$:

- A new insured starts with level 0;
- After a year without claims an insured moves up one level. If the insured is already at level 3, then the insured stays at level 3;
- After a year with a claim an insured moves down two levels. If the insured is either at level 0 or -1 , then the insured moves to level -1 ;

The probability that the insured will have a car accident in a year depends on the state, and is given by $1/2^{(\text{level}+2)}$. The insurance company models the evolution of the insured using a Markov chain.

1. Define the states of the Markov chain and give the probability transition matrix.
2. For all states, give the period, whether it is transient, recurrent, positive recurrent, and/or ergodic.
3. What is the long term proportion of insured in each of the states?
4. An insured starts on level 0, what is the expected time before the individual returns to level 0?

Mean Time Spent in Transient States

For a finite chain, let $T = \{1, \dots, t\}$ be the transient states and $\mathbf{P}_T = (P_{ij})_{i,j \in T}$. At least one row of $\mathbf{P}_T$ sums to less than 1. The shortfall in row i is the probability of moving from i to a recurrent state, from which the chain cannot return to T .

For $i, j \in T$, let s_{ij} be the expected number of periods spent in state j given $X_0 = i$. Conditioning on the first transition,

$$s_{ij} = \delta_{ij} + \sum_{k \in T} P_{ik} s_{kj}, \quad \text{where } \delta_{ij} = 1 \text{ if } i = j \text{ and } 0 \text{ otherwise.}$$

$$\text{With } \mathbf{S} = (s_{ij}) : \quad \mathbf{S} = \mathbf{I} + \mathbf{P}_T \mathbf{S} \quad \Longrightarrow \quad \boxed{\mathbf{S} = (\mathbf{I} - \mathbf{P}_T)^{-1}}$$

Interpreting $\mathbf{S}$

The period at time 0 is counted (the term δ_{ij}), so $s_{ii} \geq 1$. Row i of $\mathbf{S}$ sums to the expected time spent in T starting from i .

Example: **S** for Two Transient States

In the lecture example the transient states are 3 and 4, with

$$\mathbf{P}_T = \begin{pmatrix} 1/4 & 1/4 \\ 1/2 & 1/2 \end{pmatrix}, \quad \mathbf{I} - \mathbf{P}_T = \begin{pmatrix} 3/4 & -1/4 \\ -1/2 & 1/2 \end{pmatrix}.$$

The row for state 3 sums to $1/2$, the probability of leaving T from state 3. The row for state 4 sums to 1, so T cannot be left directly from state 4.

By the 2×2 inverse formula, with determinant $\frac{3}{4} \cdot \frac{1}{2} - \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{4}$,

$$\mathbf{S} = 4 \begin{pmatrix} 1/2 & 1/4 \\ 1/2 & 3/4 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix},$$

with rows and columns in the order 3, 4. From state 3 the chain spends on average 2 periods in state 3 and 1 in state 4, so 3 in T . From state 4 it spends 2 and 3, so 5 in T .

The lecture labels these entries by position, s_{11} , s_{12} , s_{21} , s_{22} . By state they are s_{33} , s_{34} , s_{43} , s_{44} .

Exercise 2.18: Income insurance: transient states and time spent working

Exercise 2.18: [dmc17, Solution] An insurer provides income insurance through disability insurance and unemployment insurance. To simplify the model, the insurer only consider four states, namely: employed, unemployed, disabled and retired. The yearly probability of losing a job is 2%, the yearly probability of getting a job when unemployed is 40%, the yearly probability of becoming disabled is 1% (only happens if working) and the yearly probability of recovering from disability and returning to work is 2%. The yearly probability of retiring is 2.5% when employed, 4% when unemployed, and 8% when injured.

1. Define the states of the Markov chain and give the probability transition matrix.
2. Which states are transient and which are recurrent.
3. What is the expected proportion of the working life (i.e., the time before you are not retired = working time + disability time + unemployed time) that is spent working, given the insured was initially employed? You are given that:

$$\begin{bmatrix} 0.055 & -0.02 & -0.01 \\ -0.4 & 0.44 & 0 \\ -0.02 & 0 & 0.1 \end{bmatrix}^{-1} = \begin{bmatrix} 28.7206 & 1.3055 & 2.8721 \\ 26.1097 & 3.4595 & 2.6110 \\ 5.7441 & 0.2611 & 10.5744 \end{bmatrix}$$

4. The Markov chain in this exercise assumes that the transition probabilities do not depend on age. Comment on the appropriateness of this assumption for answering question 3.

The Exponential Distribution

$X \sim \text{Exp}(\lambda)$ has density $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$, where $\lambda > 0$ is the **rate**.

Mean and variance	$E[X] = 1/\lambda, \quad \text{Var}[X] = 1/\lambda^2$
Distribution function	$F(x) = 1 - e^{-\lambda x}$
Survival function	$S(x) = 1 - F(x) = e^{-\lambda x}$
Moment generating function	$\phi(t) = E[e^{tX}] = \lambda/(\lambda - t)$ for $t < \lambda$
Hazard rate	$\mu(x) = f(x)/S(x) = \lambda$

The hazard rate is constant: given survival to x , the probability of the event in the next short interval δx is approximately $\lambda \delta x$.

Memoryless property. For all $s, t \geq 0$,

$$\Pr(X > s + t \mid X > t) = \frac{e^{-\lambda(s+t)}}{e^{-\lambda t}} = e^{-\lambda s} = \Pr(X > s),$$

so given survival to t the remaining lifetime is again $\text{Exp}(\lambda)$.

Example: the Memoryless Property

The time a customer spends in a bank queue is exponential with mean 10 minutes, so the rate is $\lambda = 1/10$ per minute.

$$\Pr(X > 15) = e^{-15/10} = e^{-1.5} = 0.223,$$

$$\Pr(X > 15 \mid X > 10) = \frac{\Pr(X > 15)}{\Pr(X > 10)} = \frac{e^{-1.5}}{e^{-1}} = e^{-0.5} = 0.607.$$

The second probability equals $\Pr(X > 5)$: given that 10 minutes have already been spent in the queue, the remaining time is again $\text{Exp}(1/10)$.

Relevance to Module 4

The exponential is the only continuous distribution with the memoryless property; its discrete analogue is the geometric. In a continuous-time Markov chain the time spent in each state is exponential, so the time already spent there gives no information about the time remaining, as the Markov property requires.

Sums of Independent Exponentials

Let $Y = X_1 + \cdots + X_n$ for independent $X_i \sim \text{Exp}(\lambda_i)$.

Equal rates, $\lambda_i = \lambda$ for all i . Then $Y \sim \text{Gamma}(n, \lambda)$, with density $f_Y(y) = \lambda e^{-\lambda y} (\lambda y)^{n-1} / (n-1)!$ for $y \geq 0$, mean n/λ and variance n/λ^2 . This follows from the m.g.f., $\phi_Y(t) = (\lambda/(\lambda - t))^n$.

Distinct rates, $\lambda_i \neq \lambda_j$ for $i \neq j$. Then

$$f_Y(y) = \sum_{i=1}^n C_{i,n} \lambda_i e^{-\lambda_i y}, \quad C_{i,n} = \prod_{j \neq i} \frac{\lambda_j}{\lambda_j - \lambda_i}.$$

The $C_{i,n}$ sum to 1 but are not all positive, so f_Y is a linear combination of the exponential densities.

Example. With $n = 2$, $\lambda_1 = 2$ and $\lambda_2 = 3$: $C_{1,2} = 3$ and $C_{2,2} = -2$, so $f_Y(y) = 6(e^{-2y} - e^{-3y})$.

The Minimum of Independent Exponentials

Let $X_{(1)} = \min(X_1, \dots, X_n)$ for independent $X_i \sim \text{Exp}(\lambda_i)$.

1. $X_{(1)} \sim \text{Exp}(\lambda)$ with $\lambda = \lambda_1 + \dots + \lambda_n$, since
 $\Pr(X_{(1)} > x) = \prod_i \Pr(X_i > x) = e^{-\lambda x}$.
2. X_i is the smallest with probability λ_i/λ .
3. Which X_i is the smallest is independent of the value of $X_{(1)}$.

Example. Two events occur after independent exponential times with rates 2 and 3 per hour. The time to the first event is $\text{Exp}(5)$, with mean $1/5$ hour (12 minutes). It is event 1 with probability $2/5$ and event 2 with probability $3/5$, whatever the time at which it occurs.

In Module 4 the X_i are the times until each possible transition out of a state: result 1 gives the time spent in the state, and result 2 the probability of each destination.

Exercise 3.1: Aggregate loss compared with premiums, by normal approximation

Exercise 3.1: [poipre1, Solution] (ACTL2131) An insurance company has a portfolio of 100 insurance contracts. The company's losses on these contracts are independent and identically distributed. Each loss X has an exponential distribution with mean 5 000 and each policyholder pays a premium of 5 050. Notice that each policyholder pays an amount larger than its expected loss. Determine the probability that the aggregate loss of the insurance company will exceed the total premiums collected. Use the normal approximation.

Exercise 3.3: Two urns: total probability and Bayes

Exercise 3.3: [poipre3, Solution] Ross: Chapter 1, Example 1.12 (modified). Consider two urns. The first contains two white and seven black balls, and the second contains five white and six black balls. We flip a fair coin and then draw a ball from the first urn or the second urn depending on whether the outcome was heads or tails.

1. What is the probability that a white ball was selected?
2. What is the conditional probability that the outcome of the toss was heads given that a white ball was selected?

Exercise 3.4: Compound sum: mean and variance of S

Exercise 3.4: [poipre4, Solution] (ACTL2131). In a portfolio of insurance policies, the amount of claim is a random variable X_k which has an exponential distribution with mean $\frac{1}{\theta}$, for $k = 1, 2, \dots$. The number of claims N in a single period is also a random variable but with a $\text{Poisson}(\lambda)$. The total claims then in the portfolio during the period is given by:

$$S = X_1 + X_2 + \dots + X_N.$$

1. Find the mean of S , $\mathbb{E}[S]$.
2. Find the variance of S , $\text{Var}(S)$.

Counting Processes

$\{N(t), t \geq 0\}$ is a **counting process** if $N(t)$ is the number of events up to time t , so that

$$N(0) = 0, \quad N(t) \geq 0, \quad N(t) \in \mathbb{Z}, \quad N(s) \leq N(t) \text{ for } s < t,$$

and $N(t) - N(s)$ is the number of events in $(s, t]$.

- ▶ **Independent increments:** the numbers of events in disjoint intervals are independent.
- ▶ **Stationary increments:** $N(t_2 + s) - N(t_1 + s)$ has the same distribution as $N(t_2) - N(t_1)$ for all $t_1 < t_2$ and $s > 0$, so the distribution depends only on the interval length.

Theorem

A counting process with independent increments is a Markov process: $N(t) - N(s)$ is independent of the history up to time s , so given $N(s)$ the earlier values give no further information about $N(t)$.

The Poisson Process

A counting process $\{N(t), t \geq 0\}$ is a **Poisson process with rate $\lambda > 0$** if

1. $N(0) = 0$;
2. it has independent increments; and
3. the number of events in any interval of length t is Poisson with mean λt : for all $s, t \geq 0$,

$$\Pr[N(t+s) - N(s) = n] = e^{-\lambda t} \frac{(\lambda t)^n}{n!}, \quad n = 0, 1, \dots$$

Condition 3 implies stationary increments, since the distribution does not depend on s . Taking $s = 0$ gives $N(t) \sim \text{Poisson}(\lambda t)$.

Units. t is measured in the time unit of λ . With $\lambda = 2$ per week, one day has mean $2 \times \frac{1}{7} = 2/7$ and three weeks has mean 6.

To decide whether a counting process is a Poisson process, check conditions 1 to 3 separately.

Example: Poisson Probabilities

Claims arrive according to a Poisson process with rate $\lambda = 2$ per week, so with t in weeks $N(t) \sim \text{Poisson}(2t)$.

Interval	t	Mean λt	Probability
One day	$1/7$	$2/7$	$\Pr(N(\frac{1}{7}) = 2) = e^{-2/7} \frac{(2/7)^2}{2!} = 0.031$
Two weeks	2	4	$\Pr(N(2) = 2) = e^{-4} \frac{4^2}{2!} = 0.147$
Three weeks	3	6	$\Pr(N(3) < 2) = e^{-6}(1 + 6) = 0.017$

In the last row, $\{N(3) < 2\} = \{N(3) = 0\} \cup \{N(3) = 1\}$, which has probability $e^{-6} + 6e^{-6}$.

By stationary increments the same probabilities apply to any interval of the same length, wherever it starts. This is not true of the non-homogeneous Poisson process later in Module 3.

Exercise 3.12: Storms in good and bad years: is it a Poisson process?

Exercise 3.12: [poi8, Solution] Ross: Chapter 5, Q86. In good years, storms occur according to a Poisson process with rate 3 per unit time, while in other years they occur according to a Poisson process with rate 5 per unit time. Suppose next year will be a good year with probability 0.3. Let $N(t)$ denote the number of storms during the first t time units of next year.

Note: We consider only $t < 1$. For the time between 0 and 1 it is either a good year or a bad year, for the time between 1 and 2 it is either a good year or a bad year, etc.

1. Find $\Pr(N(t) = n)$.
2. Does $\{N(t) : t \geq 0\}$ have stationary increments? Why or why not?
3. Does it have independent increments? Why or why not?
4. Is $\{N(t) : t \geq 0\}$ a Poisson process?
5. If next year starts off with 3 storms by time $t = 1$, what is the conditional probability it is a good year?

Common Mistakes

Not this	This
mean m taken as $\lambda = m$	$\lambda = 1/m$
a weekly rate used for one day	mean λt , with t in the units of λ
every row of $\mathbf{P}_T$ sums to 1	at least one row sums to less than 1
$s_{ij} < 1$	$s_{ij} \geq 1$: the period at time 0 counts
s_{11} taken to refer to state 1	s_{11} refers to the first state listed in T
$\sum_i X_i$ is exponential	$\min_i X_i$ is; $\sum_i X_i$ is Gamma if i.i.d.
“fewer than 2” taken as $N \leq 2$	$N = 0$ or $N = 1$

Thanks for coming!